

# Statistical Analysis Plan: ADNI MCI Conversion Prediction

PHB 243b Project 2

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# 1 Introduction

## 1.1 Background and Rationale

[Your background text here]

## 1.2 Research Questions

### Primary Research Question:

[Your primary question here]

### Secondary Questions (if any):

1. Which predictor categories (sociodemographic, clinical, neuropsychological) contribute most to prediction accuracy?
2. Does prediction performance differ between EMCI and LMCI subgroups?

## 1.3 Hypotheses

### Primary Hypothesis:

[Your hypothesis here, e.g., “Baseline ADAS-Cog-13, CDR-SB, and RAVLT forgetting scores will significantly predict MCI-to-AD conversion...”]

### Expected Performance:

Based on Grassi et al. (2019), we expect AUROC in the range of 0.75-0.85.

## 2 Data Description

### 2.1 Data Source

Data for this analysis come from the Alzheimer’s Disease Neuroimaging Initiative (ADNI), accessed via the ADNIMERGE2 R package (Donohue et al., 2026). ADNI is a longitudinal multicenter study designed to develop clinical, imaging, genetic, and biochemical biomarkers for early detection and tracking of Alzheimer’s disease.

**Data access:** [adni.loni.usc.edu](https://adni.loni.usc.edu)

**Data download date:** [Your download date]

**Citation:** Donohue MC, et al. (2026). Alzheimer’s clinical research data via R packages: The alzverse. *Alzheimer’s Dement*, 22:e71152.

### 2.2 Study Population

#### Inclusion Criteria:

1. Baseline diagnosis of Mild Cognitive Impairment (EMCI or LMCI)
2. At least 3 years of follow-up OR documented conversion to AD
3. Complete data on primary predictor variables

#### Exclusion Criteria:

1. Baseline diagnosis of CN or Dementia
2. No follow-up visits after baseline
3. Missing data on >50% of predictor variables

### 2.3 Sample Size

#### Expected Sample Size:

| Group             | N (expected) | Percentage |
|-------------------|--------------|------------|
| Stable MCI (sMCI) | ~350         | ~64%       |
| Converters (cAD)  | ~200         | ~36%       |
| <b>Total</b>      | ~550         | 100%       |

Based on Grassi et al. (2019) cohort selection criteria.

## 3 Variables

### 3.1 Primary Outcome

| Attribute          | Description                                          |
|--------------------|------------------------------------------------------|
| Variable name      | <b>converter</b>                                     |
| Definition         | Conversion from MCI to AD dementia                   |
| Type               | Binary (0 = Stable, 1 = Converter)                   |
| Operationalization | Any follow-up diagnosis of “Dementia” within 3 years |
| Reference          | Grassi et al. (2019)                                 |

## 3.2 Primary Predictors

### 3.2.1 Sociodemographic Variables

| Variable | Definition              | Type        | Rationale                     |
|----------|-------------------------|-------------|-------------------------------|
| AGE      | Age at baseline (years) | Continuous  | Age is primary AD risk factor |
| PTEDUCAT | Years of education      | Continuous  | Cognitive reserve proxy       |
| PTGENDER | Sex                     | Binary      | Sex differences in AD risk    |
| PTMARRY  | Marital status          | Categorical | Social support proxy          |

### 3.2.2 Clinical Scales

| Variable | Definition                                 | Type       | Rationale             |
|----------|--------------------------------------------|------------|-----------------------|
| CDRSB    | CDR Sum of Boxes (0-18)                    | Continuous | Clinical staging      |
| FAQ      | Functional Activities Questionnaire (0-30) | Continuous | Functional impairment |

### 3.2.3 Neuropsychological Tests

| Variable         | Definition                    | Type       | Rationale             |
|------------------|-------------------------------|------------|-----------------------|
| MMSE             | Mini-Mental State Exam (0-30) | Continuous | Global cognition      |
| ADAS13           | ADAS-Cog 13-item (0-85)       | Continuous | AD-specific cognition |
| RAVLT_immediate  | RAVLT sum of trials 1-5       | Continuous | Verbal learning       |
| RAVLT_learning   | RAVLT trial 5 - trial 1       | Continuous | Learning rate         |
| RAVLT_forgetting | RAVLT trial 5 - delayed       | Continuous | Memory retention      |
| LDELTOTAL        | Logical Memory delayed recall | Continuous | Episodic memory       |
| TRABSCOR         | Trail Making Test B (seconds) | Continuous | Executive function    |

### 3.2.4 MCI Subtype

| Variable | Definition           | Type   | Rationale    |
|----------|----------------------|--------|--------------|
| DX_BL    | Baseline MCI subtype | Binary | EMCI vs LMCI |

## 3.3 Excluded Variables

The following variable categories are intentionally **excluded** to maximize clinical translatability (Grassi et al., 2019):

- **Imaging:** MRI volumes, PET SUVRs
- **Genetic:** APOE genotype
- **Biomarkers:** CSF Abeta, tau, p-tau

**Rationale:** These measures require specialized equipment and expertise not available in most community clinical settings.

## 4 Statistical Methods

### 4.1 Descriptive Statistics

**Continuous variables:** Mean (SD), median (IQR), range

**Categorical variables:** Frequencies, percentages

**Stratification:** By conversion status (sMCI vs cAD)

**Comparison tests:**

- Continuous: t-test or Wilcoxon rank-sum
- Categorical: Chi-square or Fisher's exact

### 4.2 Missing Data

**Strategy:** [Complete case analysis / Multiple imputation / Specify]

**Justification:** [Your justification here]

### 4.3 Machine Learning Methods

#### 4.3.1 Algorithms

We will implement at least three classification algorithms:

| Algorithm                                   | R Package                 | Key Hyperparameters                                              |
|---------------------------------------------|---------------------------|------------------------------------------------------------------|
| Logistic Regression                         | <code>glm</code>          | None                                                             |
| Random Forest                               | <code>randomForest</code> | <code>ntree</code> , <code>mtry</code>                           |
| Gradient Boosting<br>[Optional: Additional] | <code>xgboost</code>      | <code>max_depth</code> , <code>eta</code> , <code>nrounds</code> |

### 4.3.2 Cross-Validation Strategy

1. **Train/Test Split:** 70% training, 30% testing
2. **Cross-Validation:** 5-fold CV within training set
3. **Site Independence:** [If feasible, split by ADNI site]

### 4.3.3 Hyperparameter Tuning

| Algorithm     | Tuning Method | Grid/Range                                                |
|---------------|---------------|-----------------------------------------------------------|
| Random Forest | Grid search   | <code>mtry</code> : 2-8, <code>ntree</code> : 100-500     |
| XGBoost       | Grid search   | <code>max_depth</code> : 3-6, <code>eta</code> : 0.01-0.3 |

## 4.4 Evaluation Metrics

| Metric      | Definition           | Target |
|-------------|----------------------|--------|
| AUROC       | Area under ROC curve | > 0.75 |
| Sensitivity | $TP / (TP + FN)$     | > 0.70 |
| Specificity | $TN / (TN + FP)$     | > 0.70 |
| PPV         | $TP / (TP + FP)$     | Report |
| NPV         | $TN / (TN + FN)$     | Report |

**Threshold Selection:** Youden’s J statistic (maximize sensitivity + specificity)

## 4.5 Feature Importance

- **Random Forest:** Mean decrease in Gini / Permutation importance
- **Logistic Regression:** Standardized coefficients
- **XGBoost:** Gain-based importance

# 5 Planned Results

## 5.1 Table 1: Baseline Characteristics by Conversion Status

| Characteristic              | Stable MCI | Converters | p-value |
|-----------------------------|------------|------------|---------|
| N                           |            |            |         |
| Age, mean (SD)              |            |            |         |
| Education, mean (SD)        |            |            |         |
| Female, n (%)               |            |            |         |
| EMCI, n (%)                 |            |            |         |
| MMSE, mean (SD)             |            |            |         |
| ADAS-Cog-13, mean (SD)      |            |            |         |
| CDR-SB, mean (SD)           |            |            |         |
| FAQ, mean (SD)              |            |            |         |
| RAVLT immediate, mean (SD)  |            |            |         |
| RAVLT forgetting, mean (SD) |            |            |         |
| LDELTOTAL, mean (SD)        |            |            |         |
| TMT-B, mean (SD)            |            |            |         |

## 5.2 Table 2: Model Performance Comparison

| Metric         | Logistic Reg | Random Forest | XGBoost |
|----------------|--------------|---------------|---------|
| AUROC (95% CI) |              |               |         |
| Sensitivity    |              |               |         |
| Specificity    |              |               |         |
| PPV            |              |               |         |
| NPV            |              |               |         |
| Accuracy       |              |               |         |

## 5.3 Table 3: Feature Importance Rankings

| Rank | Logistic Reg | Random Forest | XGBoost |
|------|--------------|---------------|---------|
| 1    |              |               |         |
| 2    |              |               |         |
| 3    |              |               |         |
| 4    |              |               |         |
| 5    |              |               |         |

## 5.4 Planned Figures

### 5.4.1 Figure 1: Cohort Flow Diagram

- **Type:** CONSORT-style flow diagram
- **Purpose:** Show sample selection and exclusions

### 5.4.2 Figure 2: ROC Curves

- **Type:** Overlaid ROC curves for all models
- **Purpose:** Compare discrimination across algorithms

### 5.4.3 Figure 3: Feature Importance Plot

- **Type:** Bar chart or dot plot
- **Purpose:** Visualize top predictors

### 5.4.4 Figure 4: Calibration Plot

- **Type:** Predicted vs. observed probabilities
- **Purpose:** Assess calibration of best model

## 6 Sensitivity Analyses

| Analysis               | Rationale                | Expected Impact          |
|------------------------|--------------------------|--------------------------|
| EMCI only              | Test in early-stage MCI  | Lower conversion rate    |
| LMCI only              | Test in late-stage MCI   | Higher conversion rate   |
| Exclude outliers       | Robustness check         | Minimal change expected  |
| Alternative imputation | Compare to complete case | Similar results expected |
| Different CV folds     | Check stability          | Similar AUROC expected   |

## 7 Limitations

### 7.1 Scope and Generalizability

- Results specific to ADNI population (highly educated, mostly White)
- May not generalize to community clinic populations
- ADNI uses strict diagnostic criteria

### 7.2 Methodological Limitations

- Retrospective cohort design
- Cannot establish causation
- 3-year prediction horizon may miss later converters
- No external validation cohort

### 7.3 Data Limitations

- Missing data on some predictor variables
- Potential selection bias in ADNI enrollment
- Measurement variability across sites

## 8 Timeline

| Week          | Task                                | Deliverable         |
|---------------|-------------------------------------|---------------------|
| 1 (Feb 17-21) | Data preparation, cohort selection  | Clean dataset       |
| 2 (Feb 24-28) | Feature engineering, initial models | Preliminary results |
| 3 (Mar 2-6)   | Model tuning, evaluation, report    | Final submission    |

**Due Date:** March 3, 2026

## 9 References

1. Donohue MC, et al. (2026). Alzheimer’s clinical research data via R packages: The alzverse. *Alzheimer’s Dement*, 22:e71152.
2. Grassi M, et al. (2019). A Clinically-Translatable Machine Learning Algorithm for the Prediction of Alzheimer’s Disease Conversion. *Front Neurol*, 10:756.
3. Petersen RC. (2004). Mild cognitive impairment as a diagnostic entity. *J Intern Med*, 256(3):183-194.
4. Rye I, et al. (2022). Predicting conversion to Alzheimer’s disease in individuals with Mild Cognitive Impairment. *Sci Rep*, 12:15566.

## 10 Appendix: Code Framework

```
# =====  
# Data Preparation  
# =====  
library(ADNIMERGE2)  
library(tidyverse)  
library(caret)  
  
# Load data
```

```

data(ADSL)
data(ADQS)

# Select MCI subjects at baseline
mci_baseline <- ADSL |>
  filter(DX_BL %in% c("EMCI", "LMCI"))

# Define conversion status
# [Your code here]

# =====
# Feature Engineering
# =====
# Select predictor variables
predictors <- c("AGE", "PTEDUCAT", "PTGENDER", "CDRSB", "FAQ",
               "MMSE", "ADAS13", "RAVLT_immediate", "RAVLT_forgetting",
               "LDELTOTAL", "TRABSCOR")

# =====
# Model Training
# =====
# Train/test split
set.seed(243)
train_idx <- createDataPartition(mci_data$converter, p = 0.7, list = FALSE)

# Cross-validation setup
ctrl <- trainControl(method = "cv", number = 5,
                     classProbs = TRUE, summaryFunction = twoClassSummary)

# =====
# Model Evaluation
# =====
# ROC curves
library(pROC)
# [Your code here]

```